

Nose, Nasal cavity & Paranasal Sinuses

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Objectives

- *At the end of the lecture, the students should be able to:*
- Describe the boundaries of the nasal cavity.
- Describe the nasal conchae and meati.
- Demonstrate the openings in each meatus.
- Describe the paranasal sinuses and their functions

WHAT IS RESPIRATION?

Respiration is the exchange of gases between the atmosphere, the blood, and the body cells.

Three basic processes:

- 1) pulmonary ventilation.
- 2) external respiration.
- 3) internal respiration.

FUNCTIONS:

- 1) Gas exchange.
- 2) Receptors for smell.
- 3) Filter, warm, and moisten incoming air.
- 4) Produce sounds (phonation).
- 5) Eliminate some wastes other than CO_2 .

STRUCTURES: -

Upper respiratory system

Nose

Nasal cavity

Paranasal sinuses

Pharynx

Lower respiratory system

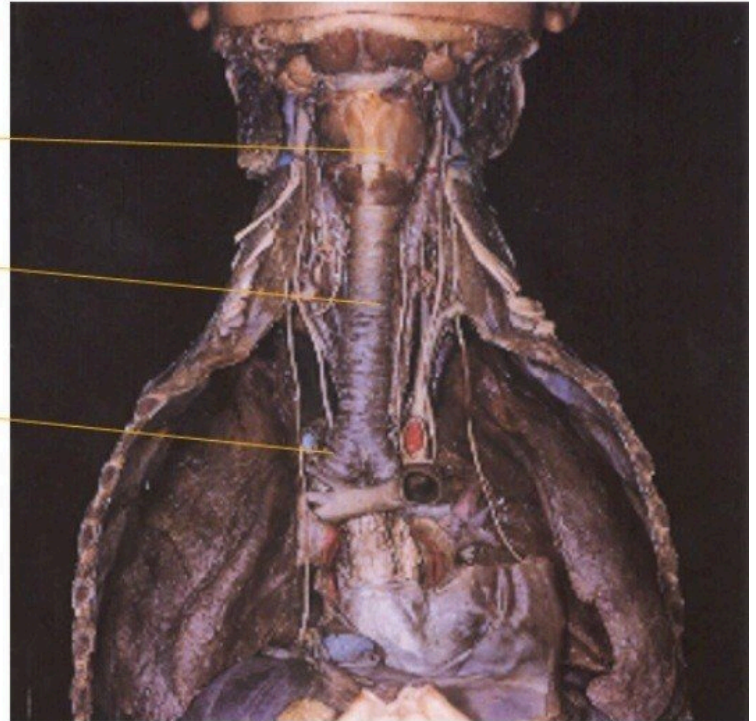
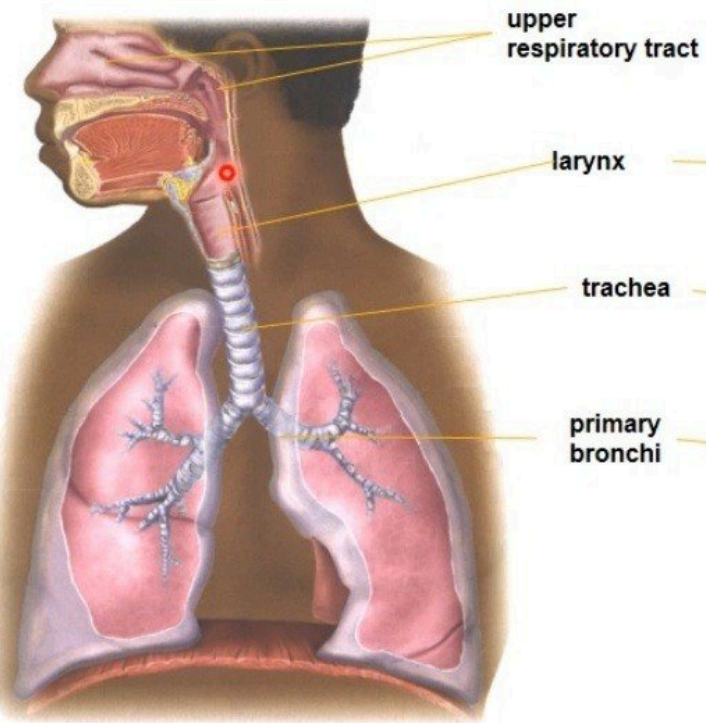
Larynx

Trachea

Bronchi

Lungs

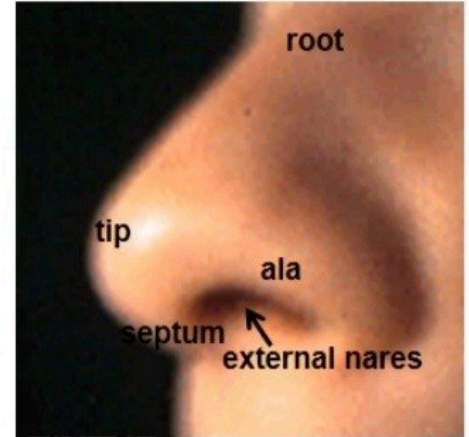
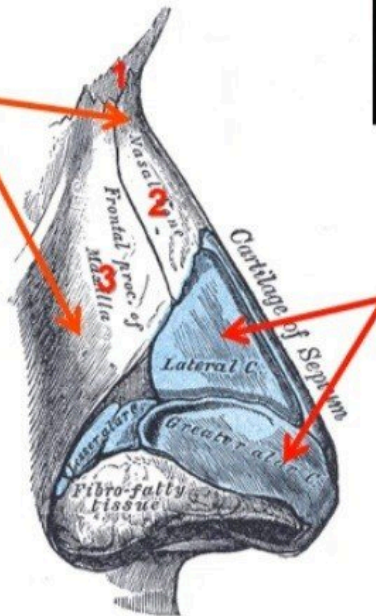
RESPIRATORY ORGANS



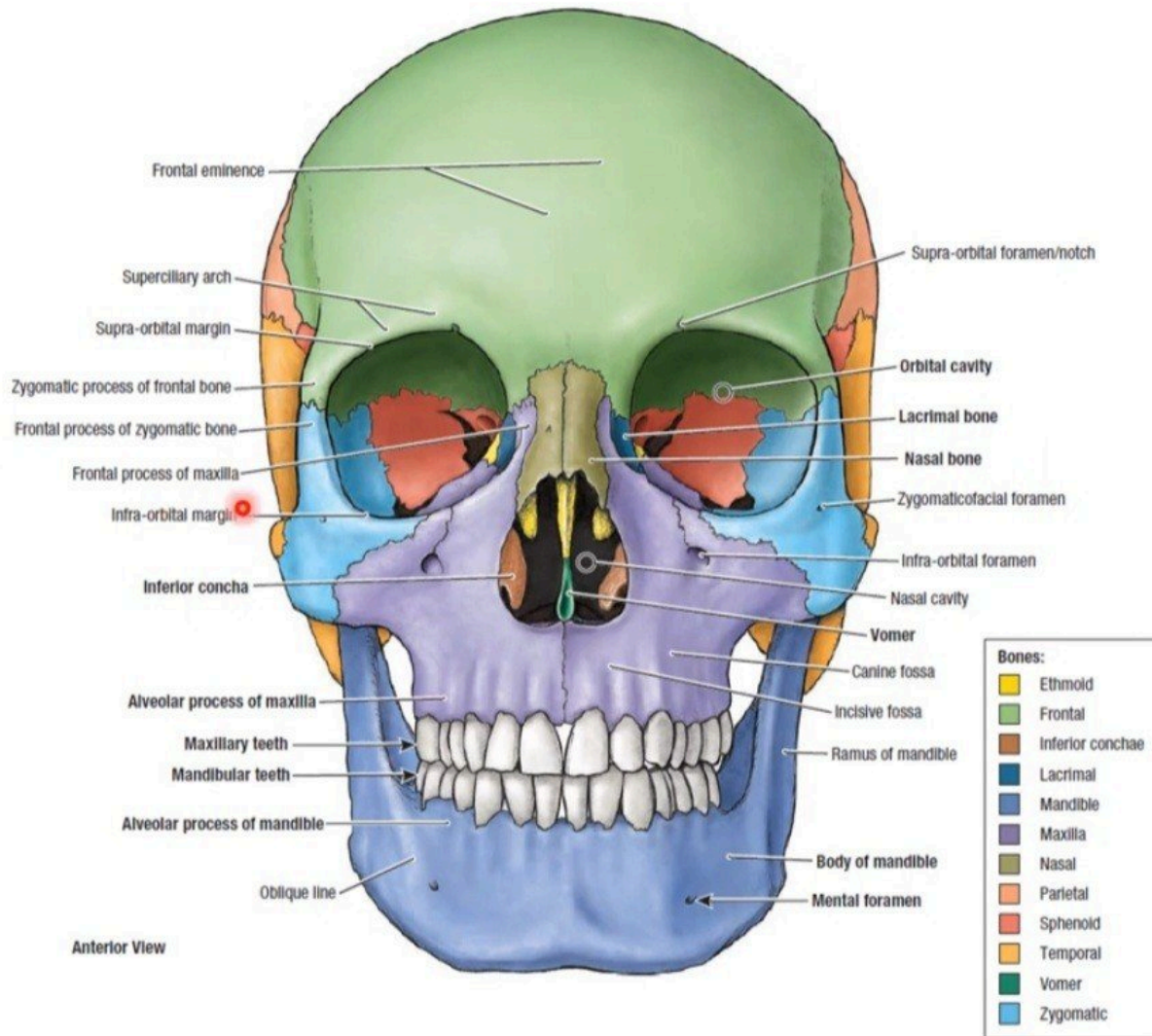
Nose

- The **external (anterior) nares or nostrils**, lead to the **nasal cavity**.

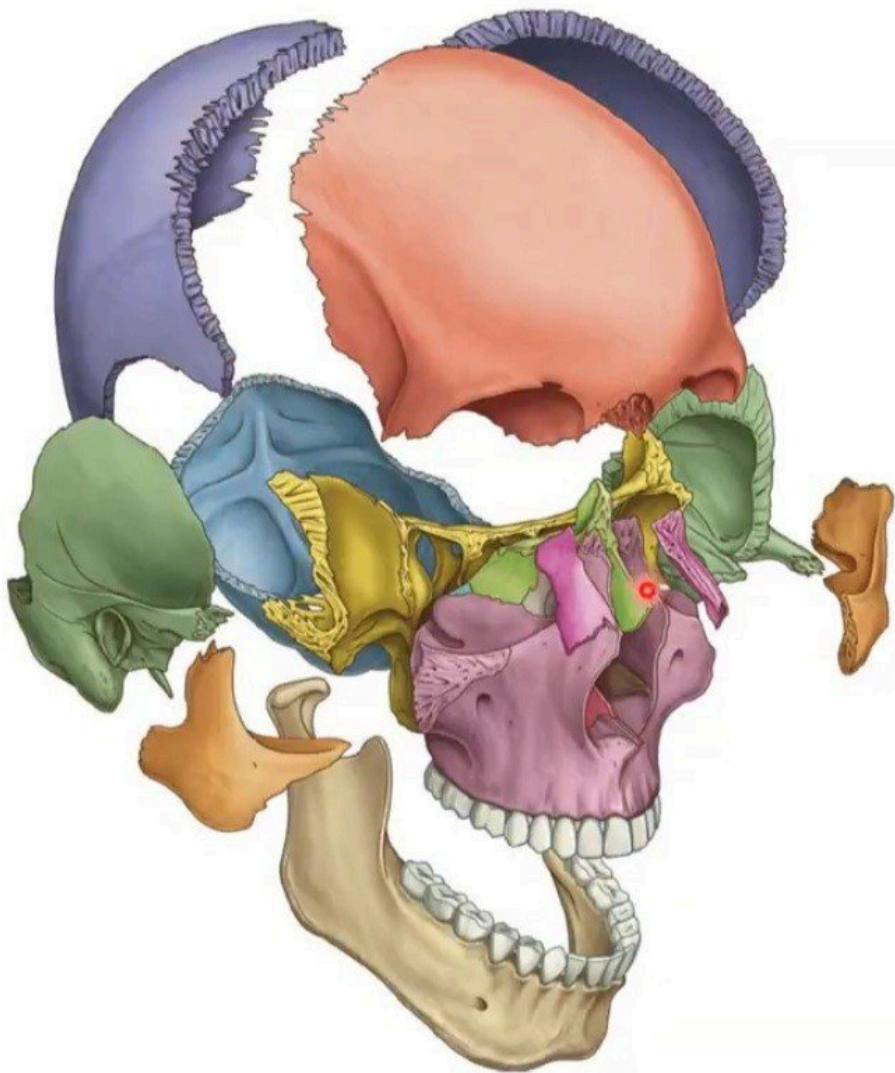
Formed above by:
Bony skeleton

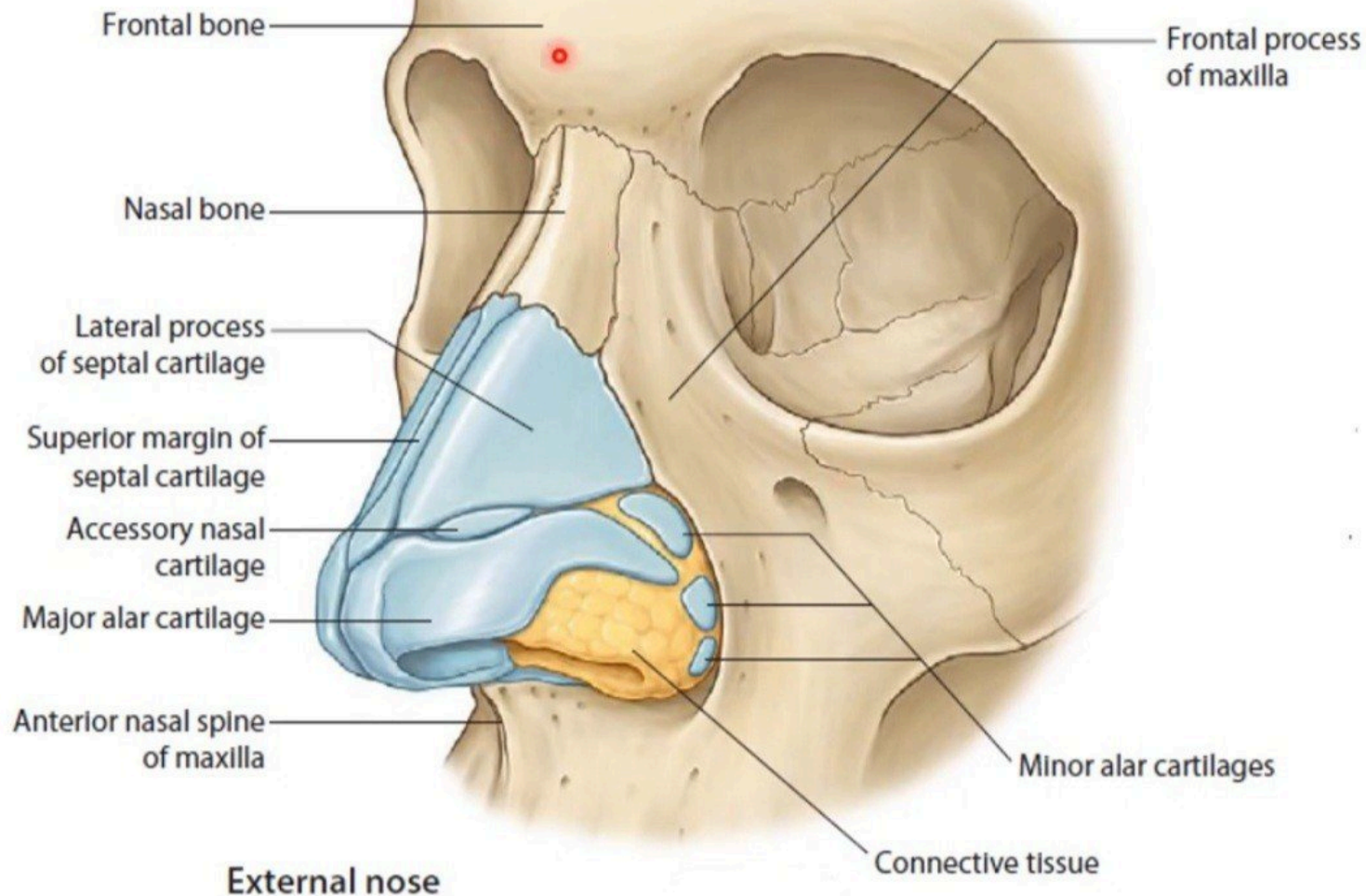


- Formed **below** by **plates of hyaline cartilage**.



Anterior View





External nose

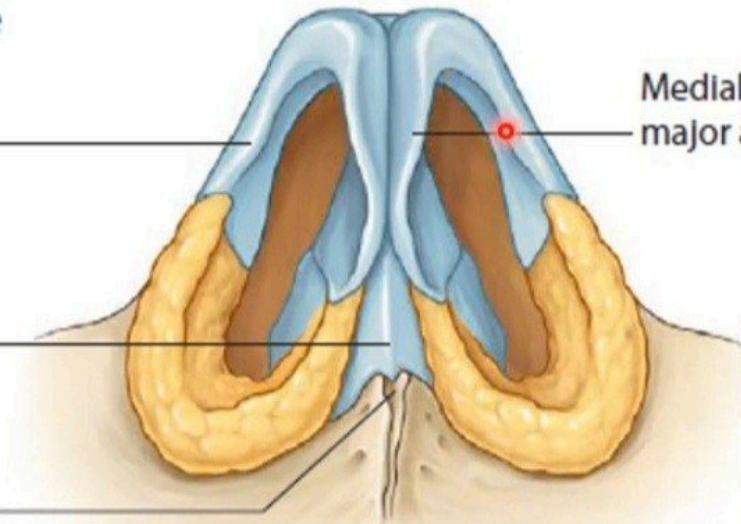
Lateral crus of
major alar cartilage

Medial crus of
major alar cartilage

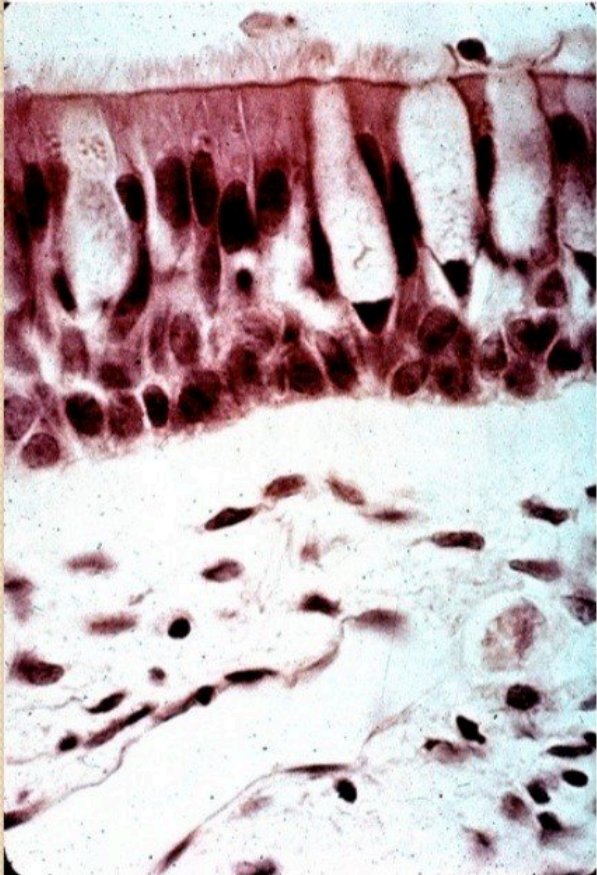
Septal nasal cartilage

Anterior nasal spine of maxilla

Inferior view



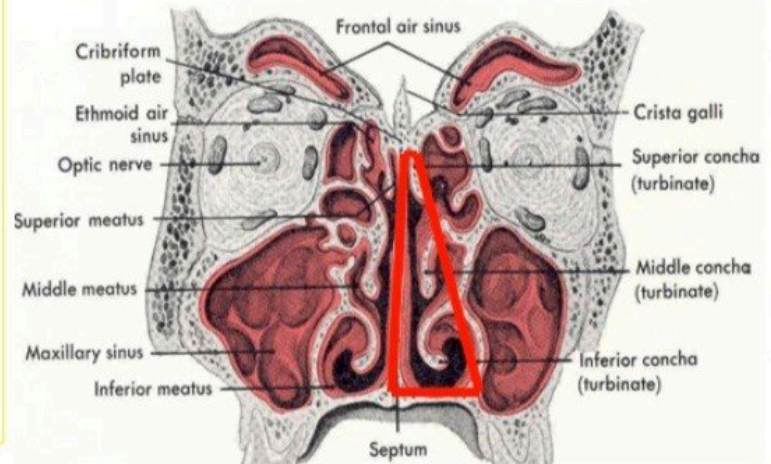
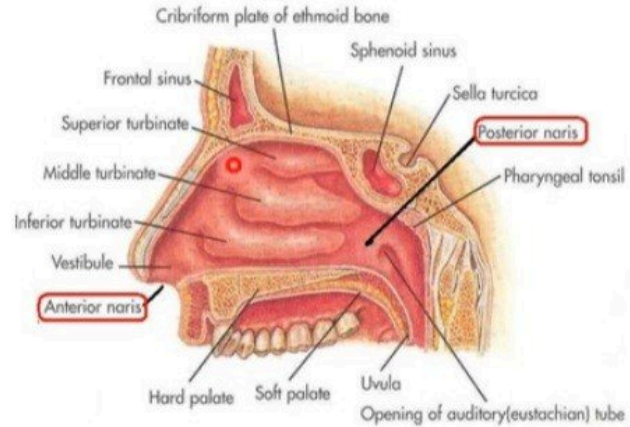
Respiration begins in the nose

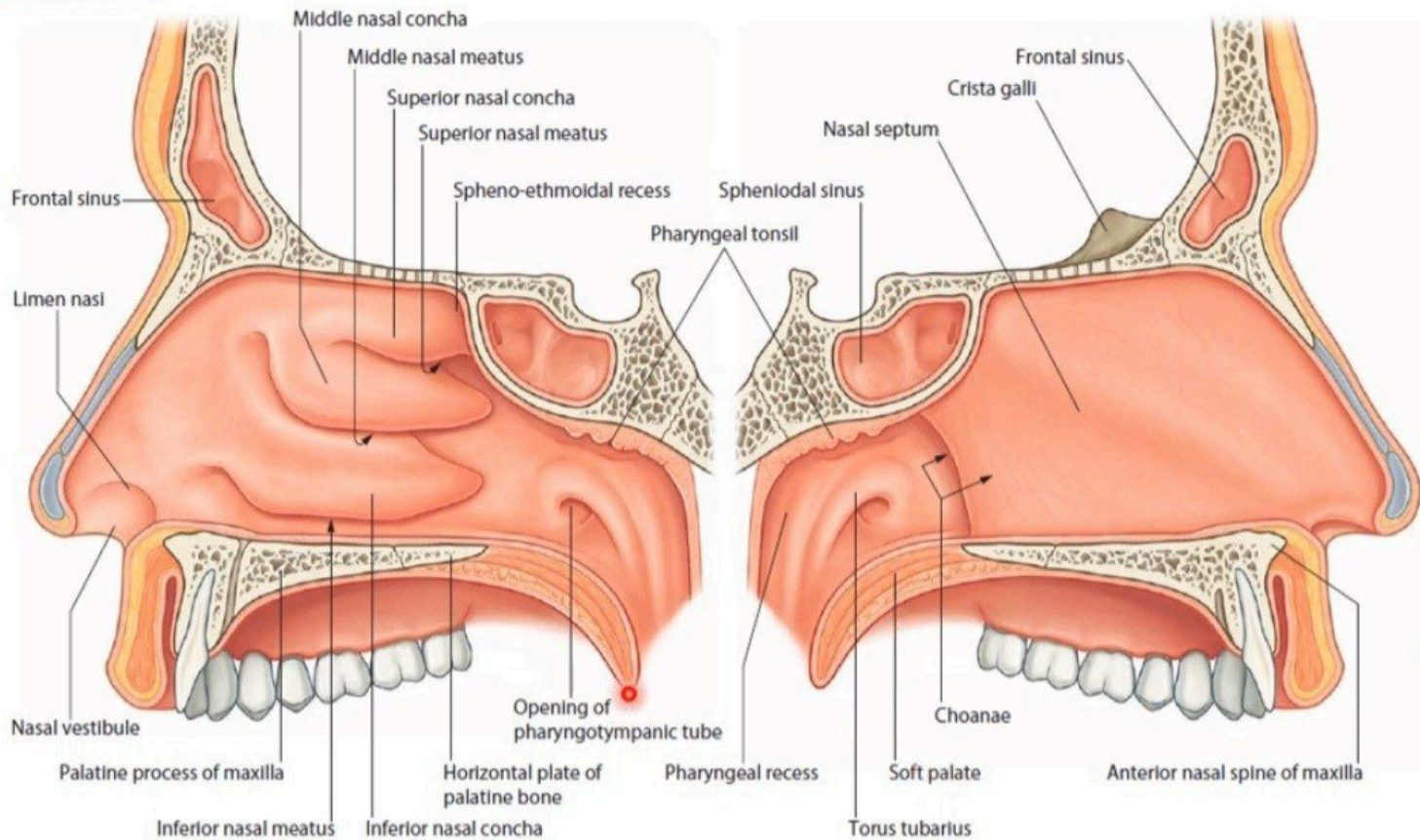


- Ciliated columnar epithelial tissue lines the nose to warm and moisten the inhaled air.
- Mucous cells are found within the lining to allow moistening of the air.

Nasal Cavity

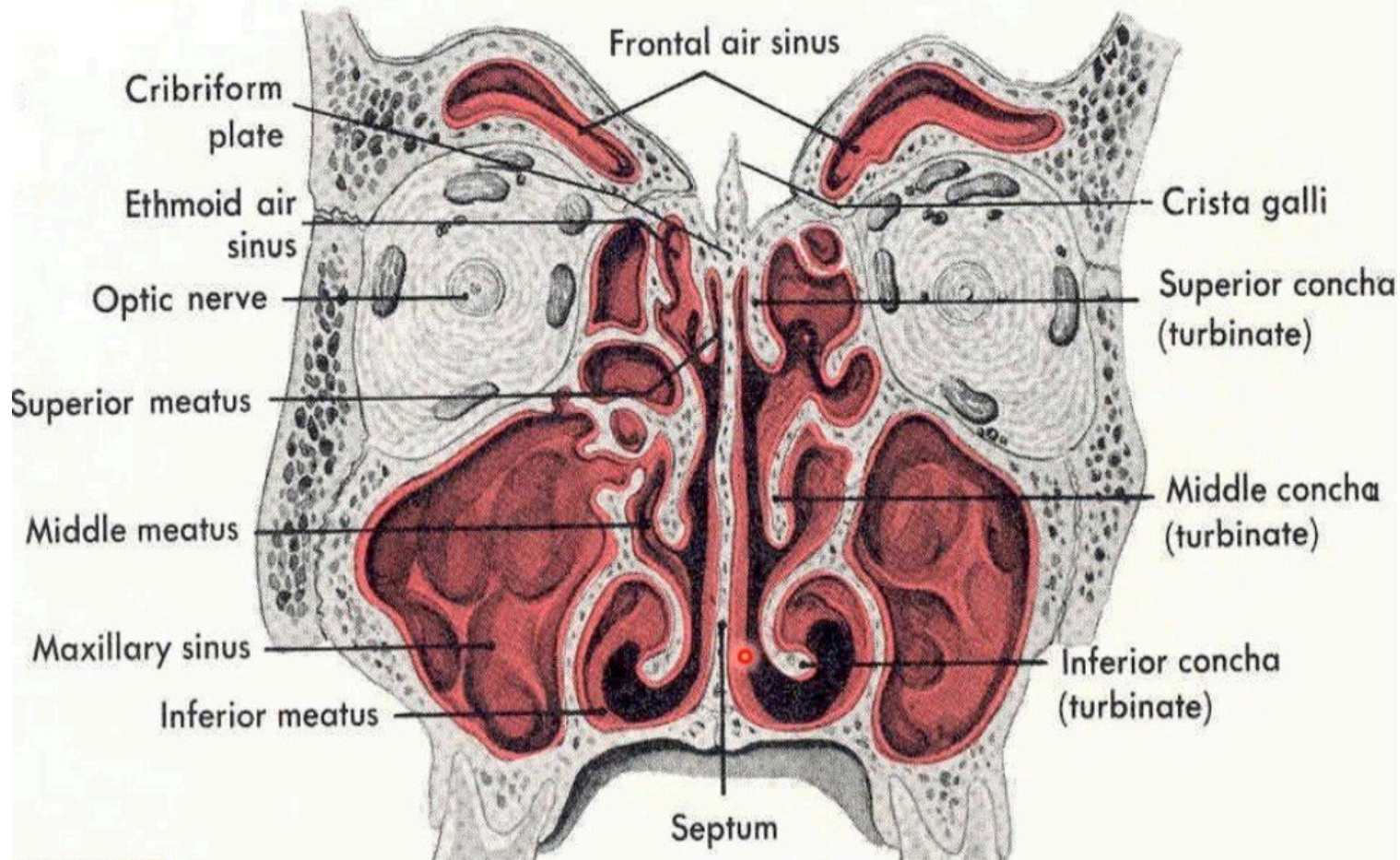
- Extends from the external (anterior) nares to the posterior nares (choanae).
- Divided into right & left halves by the nasal septum.
- Each half has a:
 - Roof
 - Lateral wall
 - Medial wall (septum)
 - Floor

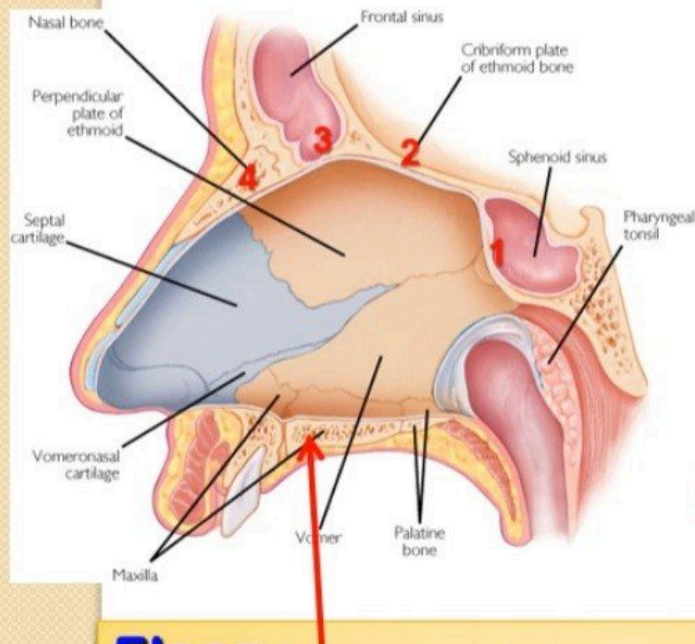




Lateral wall of the nasal cavity

Medial wall of the nasal cavity



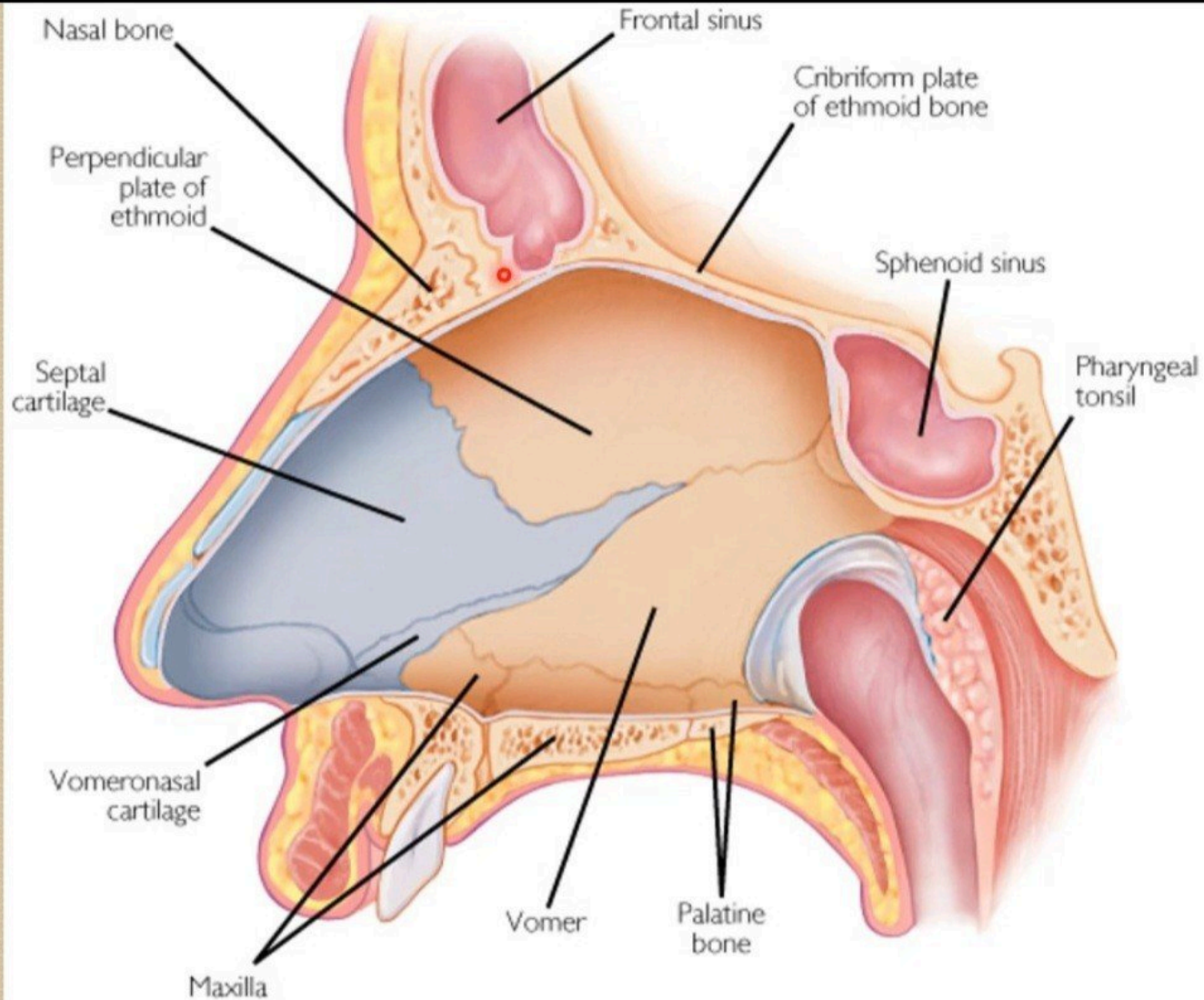


Roof

- Narrow & formed (from behind forward) by the:
 - Body of sphenoid.
 - Cribriform plate of ethmoid bone.
 - Frontal bone.
 - Nasal bone & cartilage

Floor

- Separates it from the oral cavity.
- Formed by the **hard (bony) palate**.

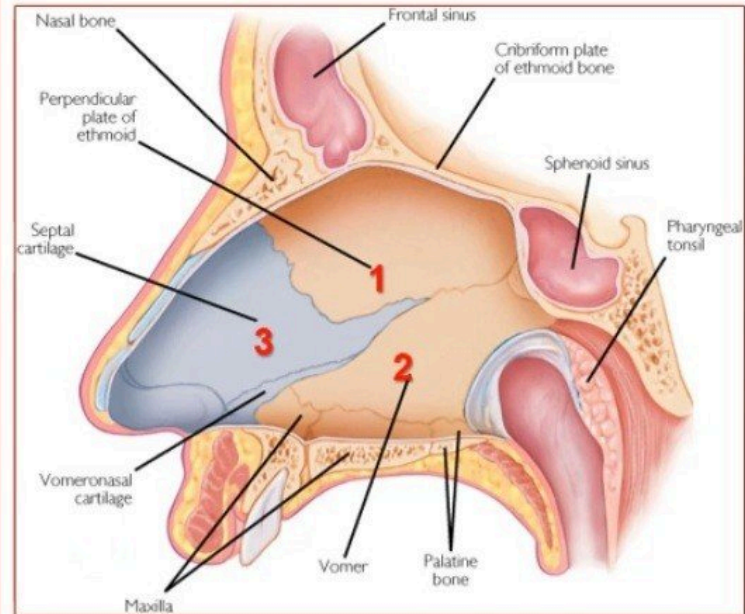


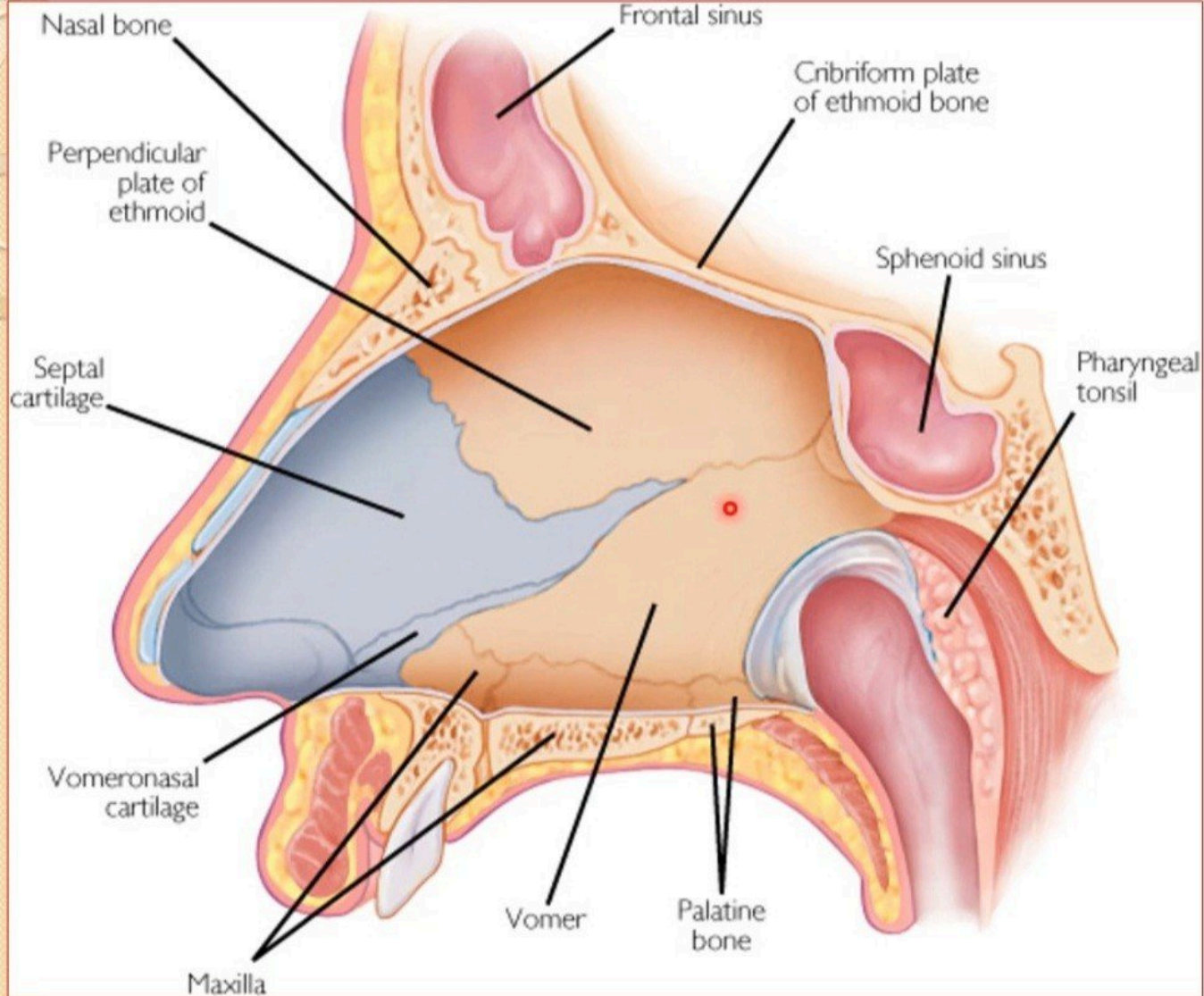
- **Medial Wall (Nasal Septum)**

- **Osteocartilaginous partition.**

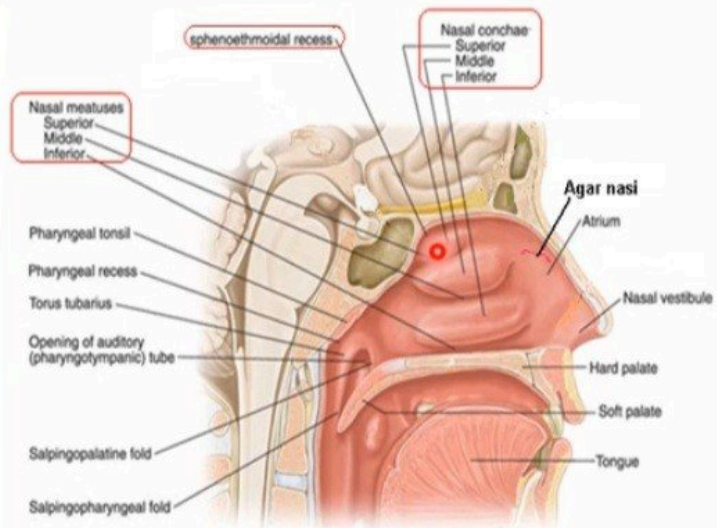
- **Formed by:**

- 1. Perpendicular plate of ethmoid bone.**
- 2. Vomer.**
- 3. Septal cartilage.**





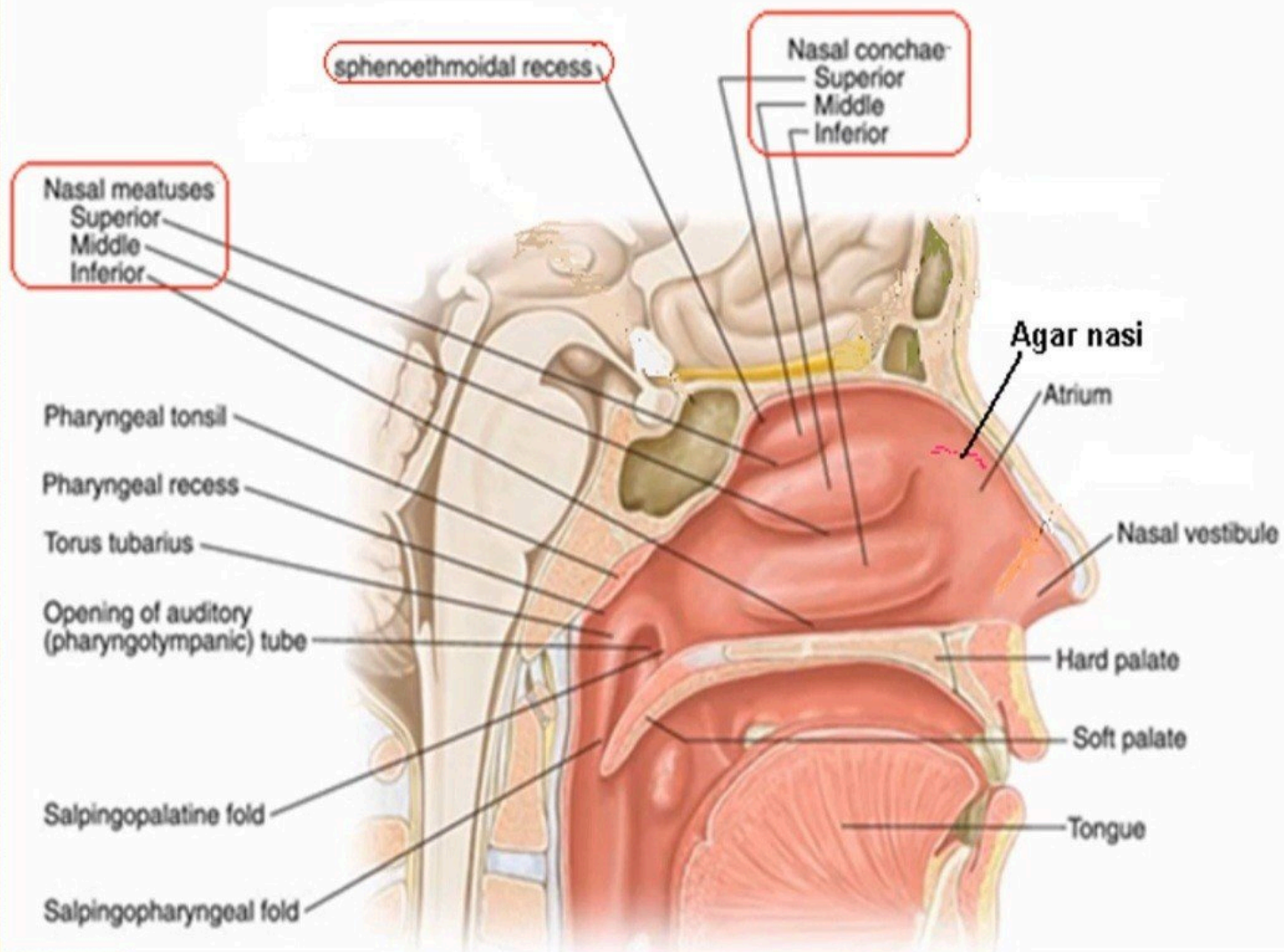
- **Lateral Wall**
- Shows three horizontal bony projections, the **superior, middle & inferior conchae**
- The cavity below each concha is called a **meatus** and are named as **superior, middle & inferior** corresponding to the conchae.



- The small space above the superior concha is the **sphenoethmoidal recess**.

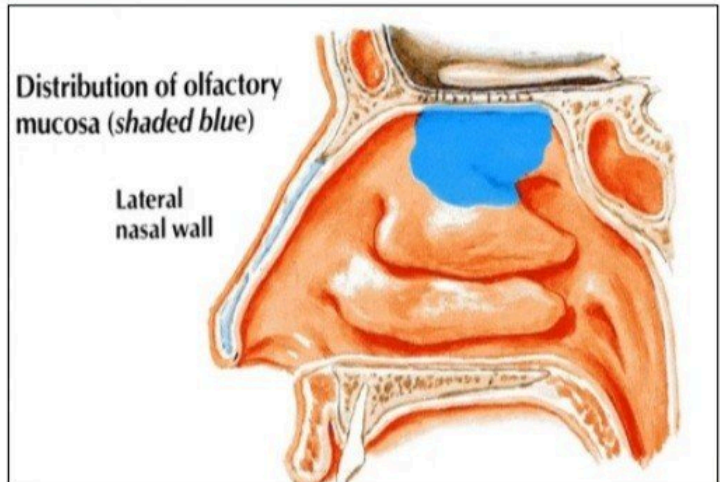
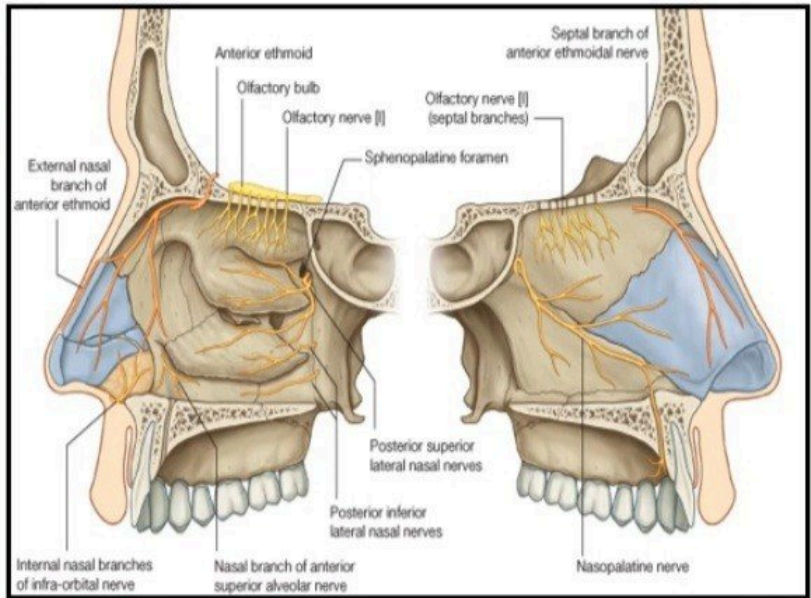
The **conchae** increase the surface area of the nasal cavity. The **recess & meati** receive the openings of the:

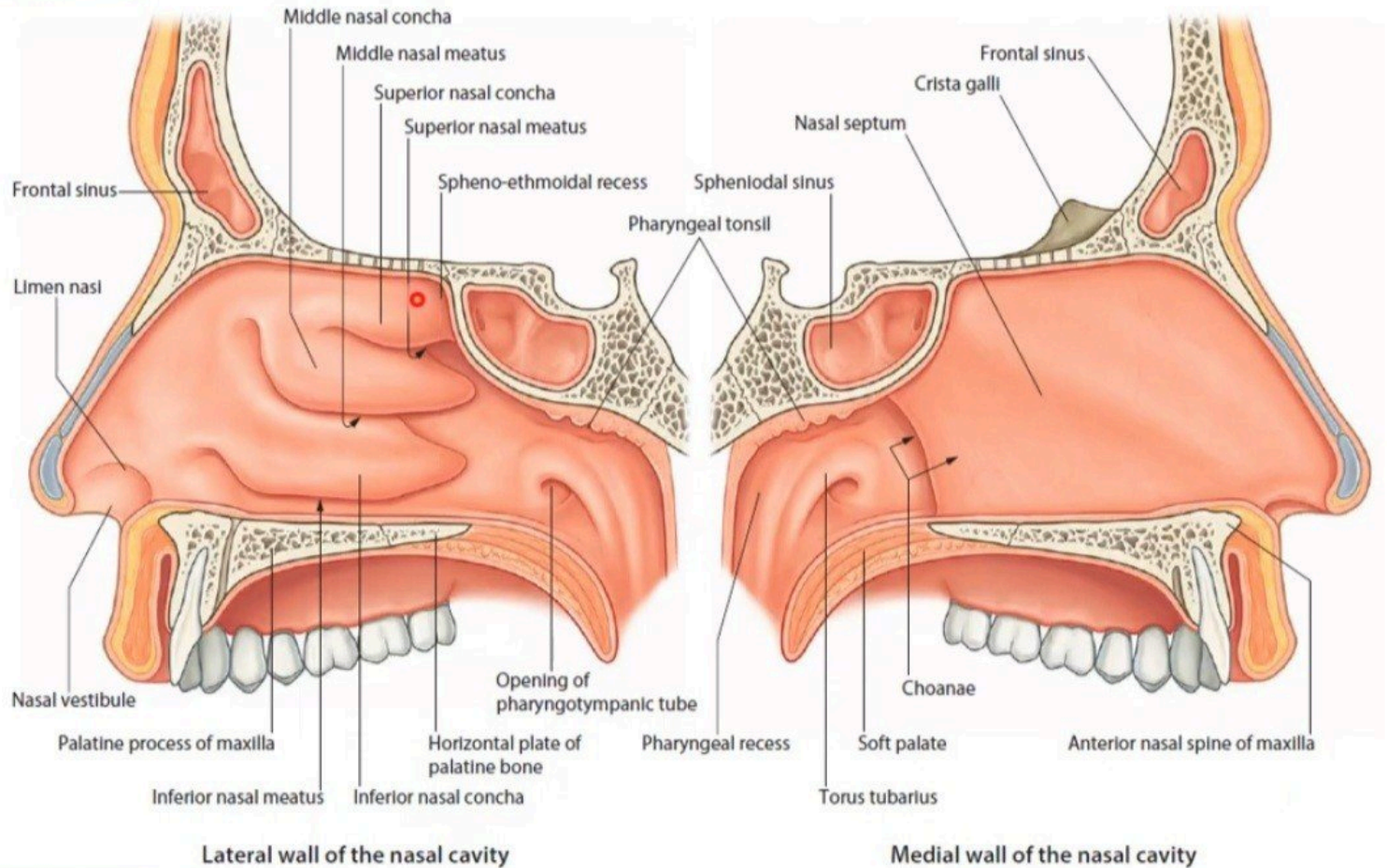
- **Paranasal sinuses.**
- **Nasolacrimal duct.**

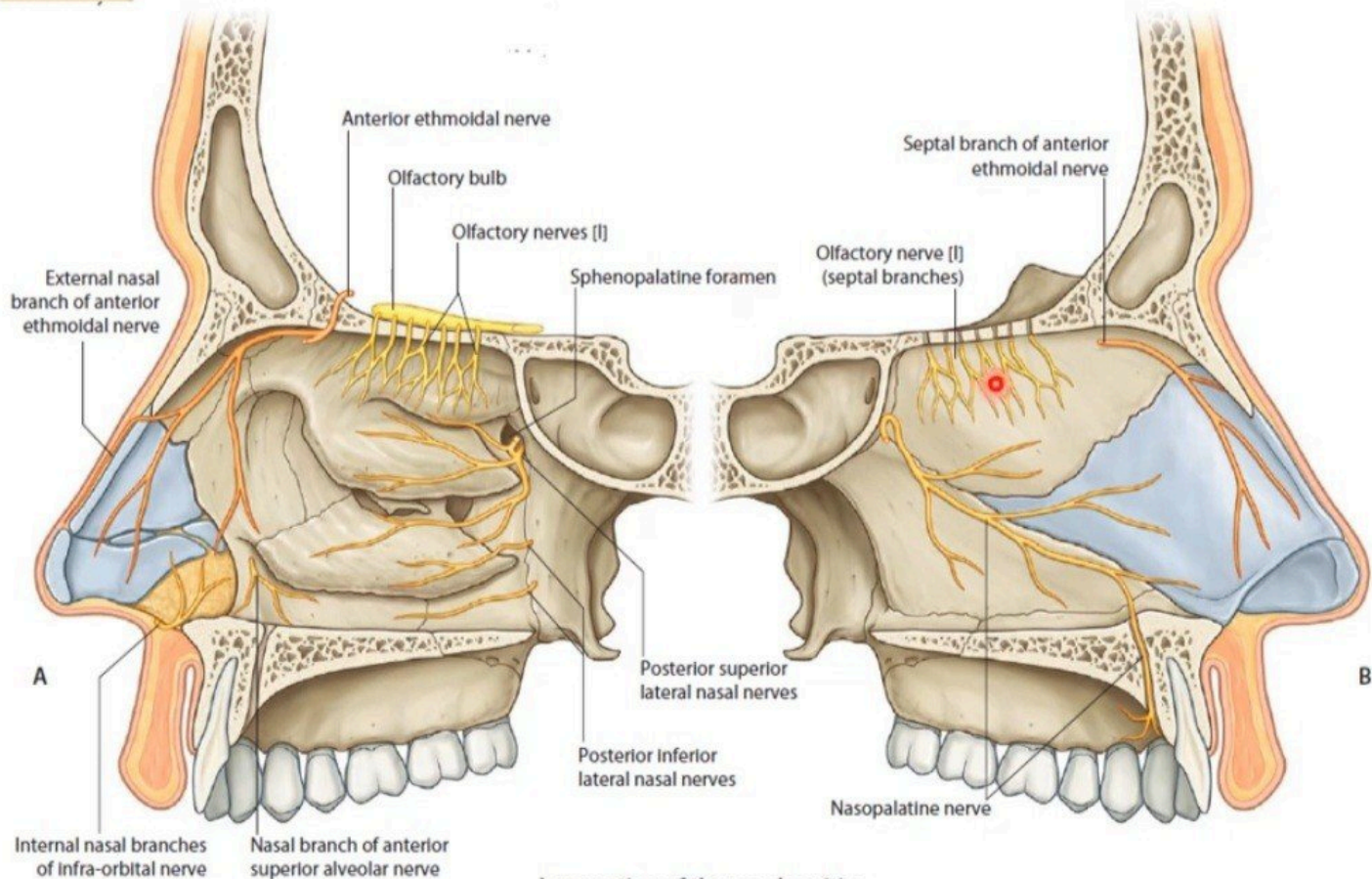


Nasal mucosa

- **Olfactory** :
- It is delicate and contains olfactory nerve cells.
- It is present in the upper part of nasal cavity:
- **Roof**,
- **On the lateral wall**,
- ◉ it lines the upper surface of the superior concha and the sphenoethmoidal recess.
- ◉ **On the medial wall**, it lines the superior part of the nasal septum.

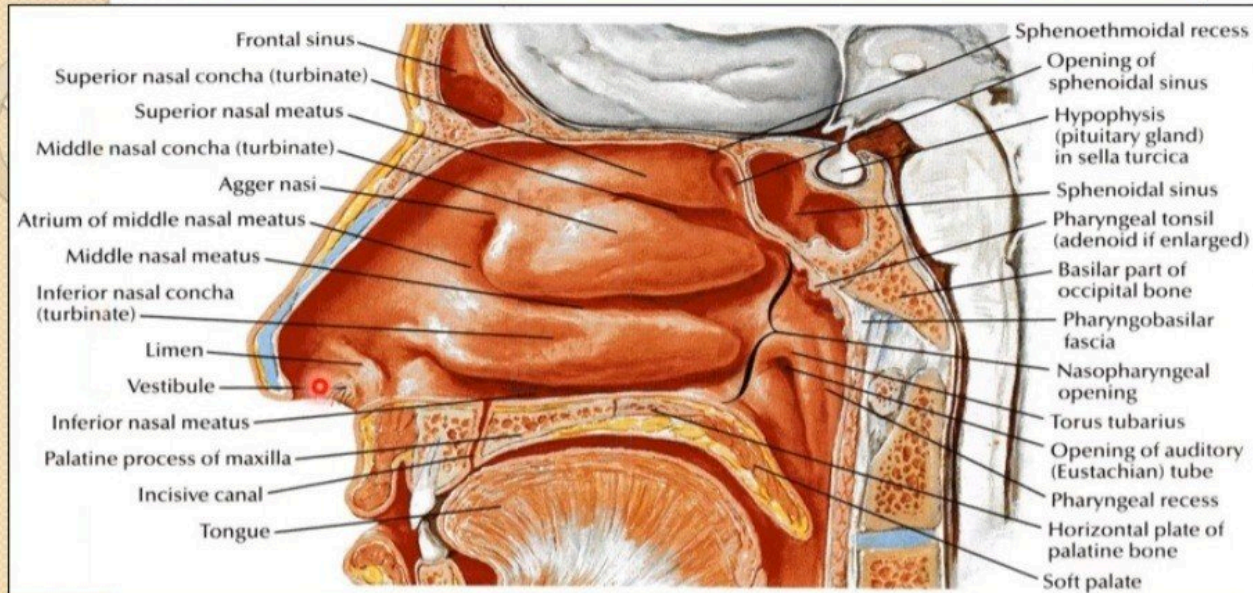




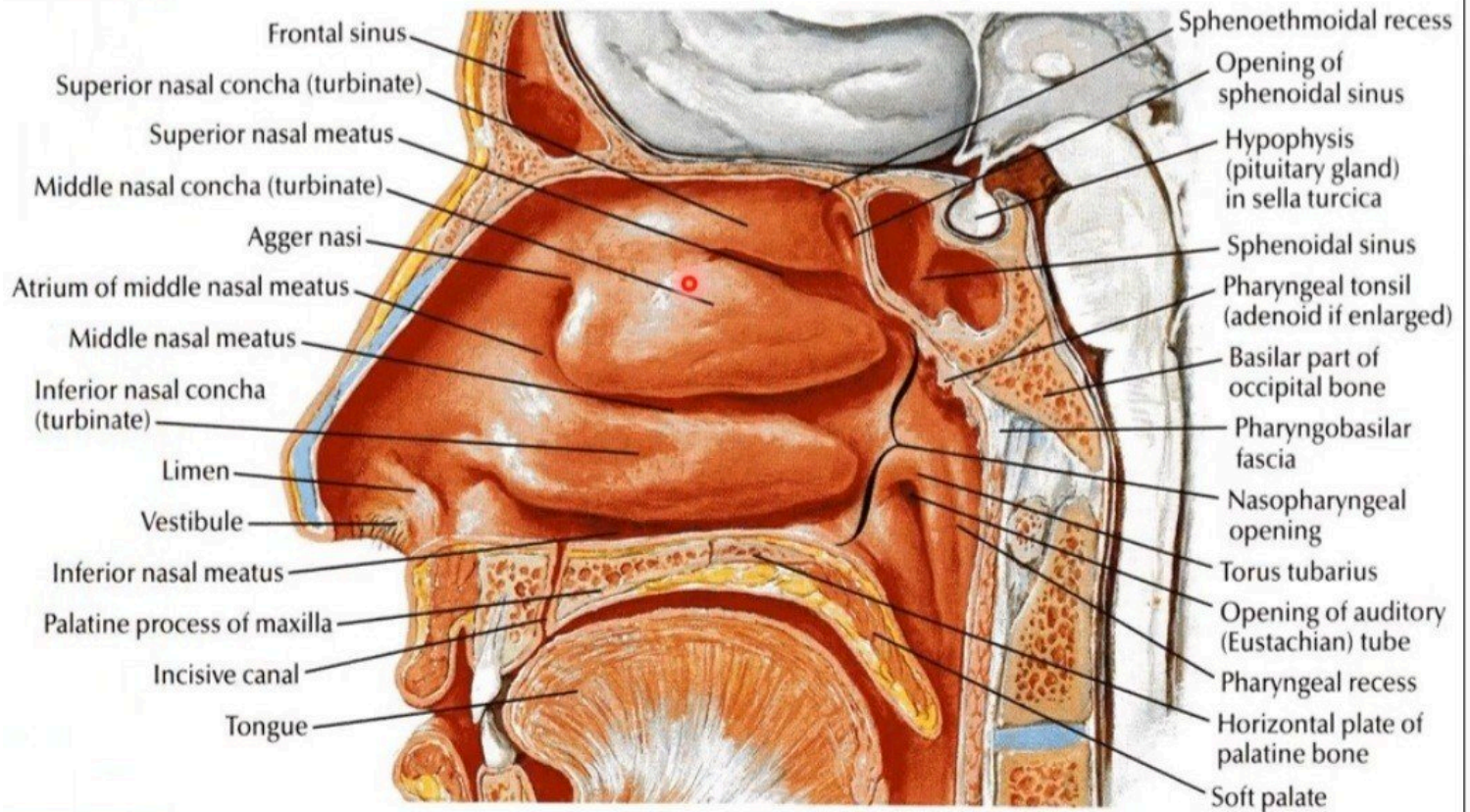


Innervation of the nasal cavities
A. Lateral wall of right nasal cavity B. Medial wall of right nasal cavity

RESPIRATORY MUCOSA

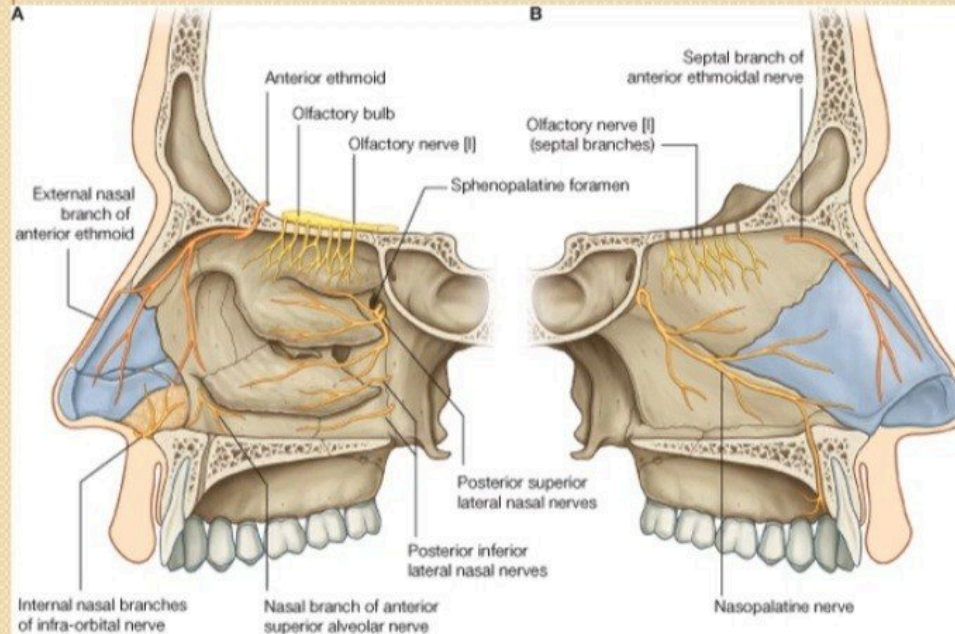


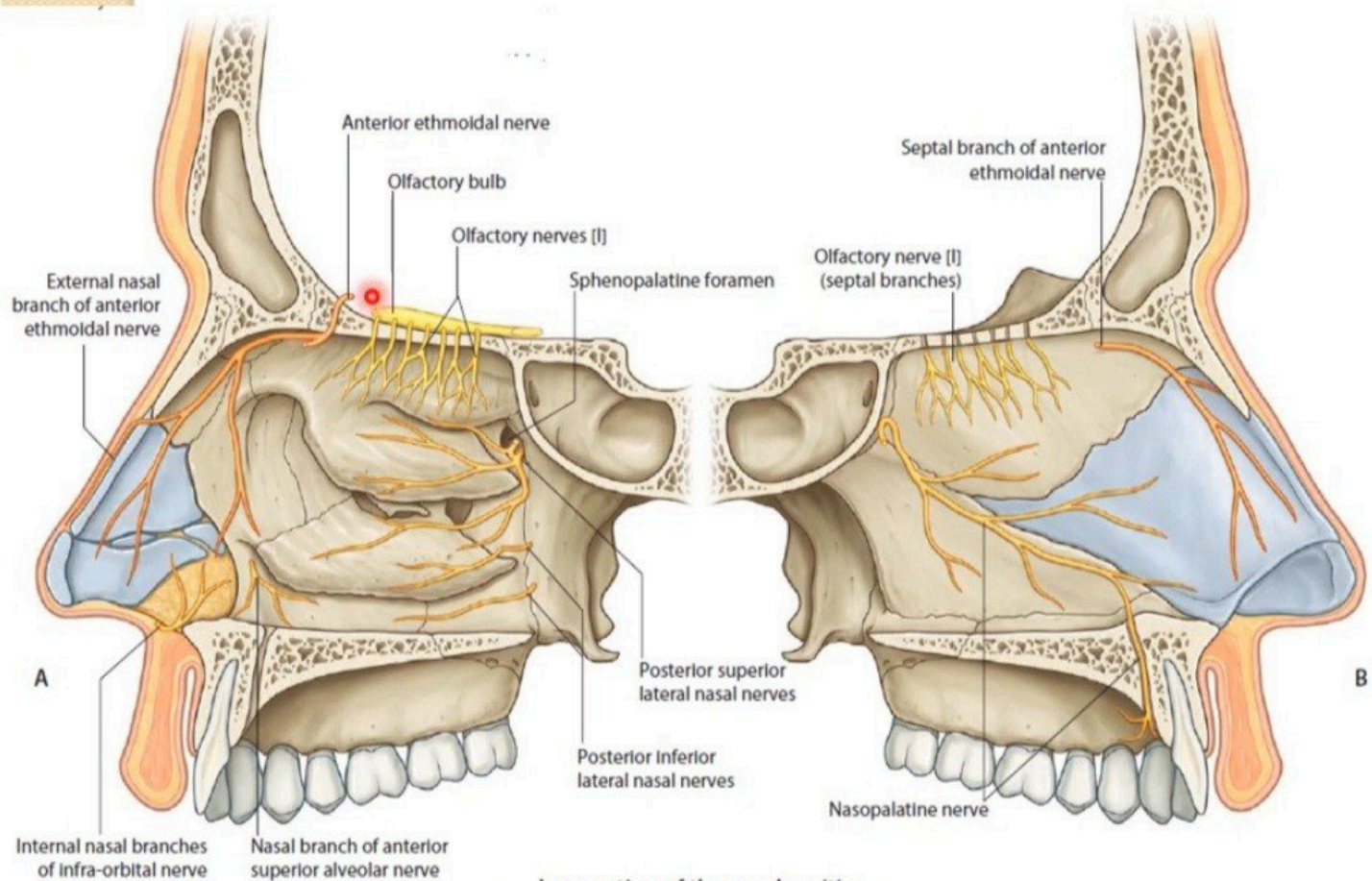
- It is **thick**, ciliated highly vascular and contains mucous glands & goblet cells
- It lines the **Lower part** of the nasal cavity.
- It functions to **moisten**, **clean** and **warm** the inspired air.
- The air is **moistened** by the secretion of numerous serous glands.
- It is **cleaned** by the removal of the dust particles by the ciliary action of the columnar ciliated epithelium that covers the mucosa.
- The air is **warmed** by a **submucous venous plexus**.
- **The Vestibule is lined by Skin.**



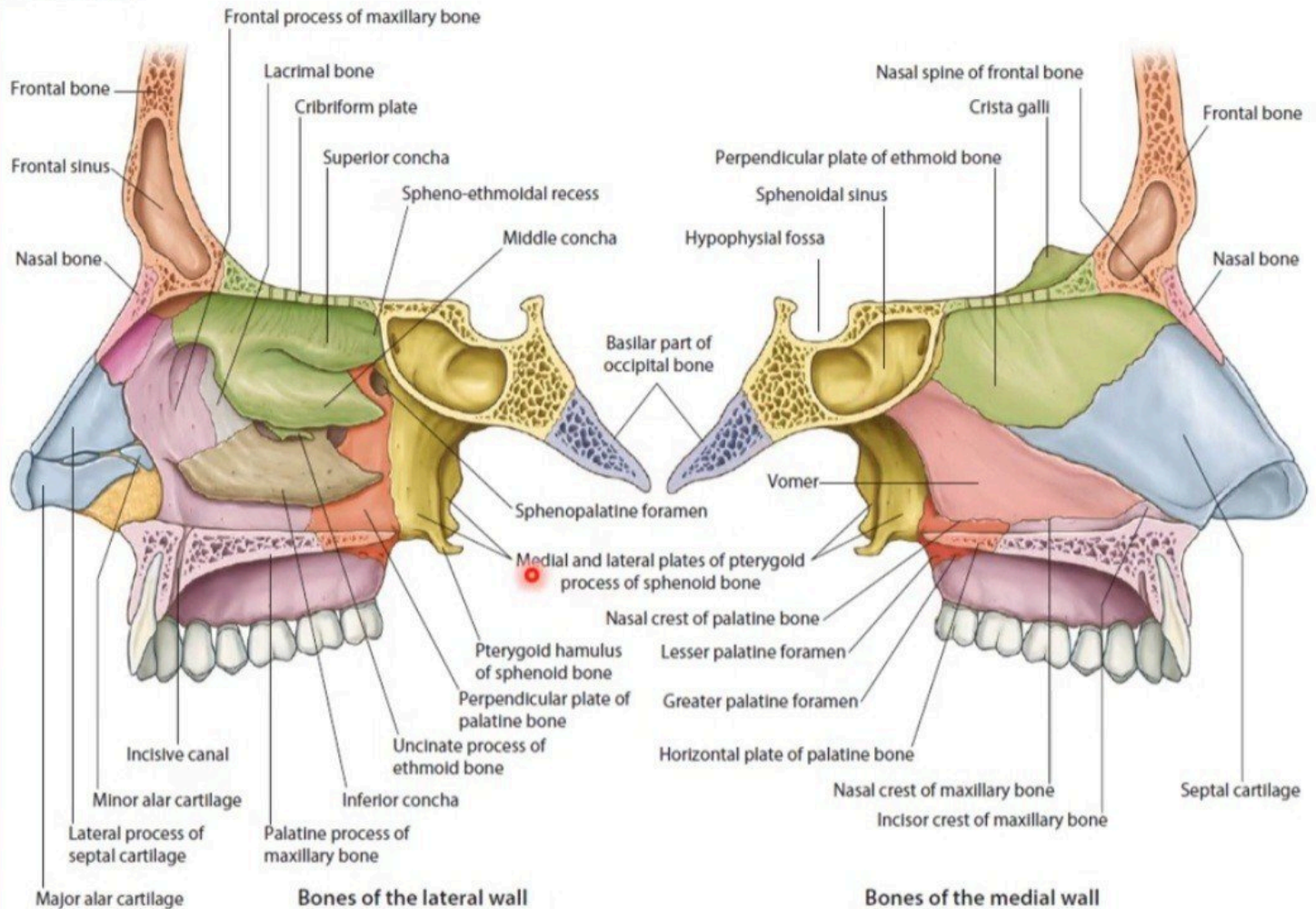
NERVE SUPPLY

- Olfactory mucosa supplied by **olfactory nerves**.
- Nerves of general sensation are derived from
 - **ophthalmic**
 - **maxillary nerves.**
 - **Autonomic fibers.**





Innervation of the nasal cavities
A. Lateral wall of right nasal cavity B. Medial wall of right nasal cavity



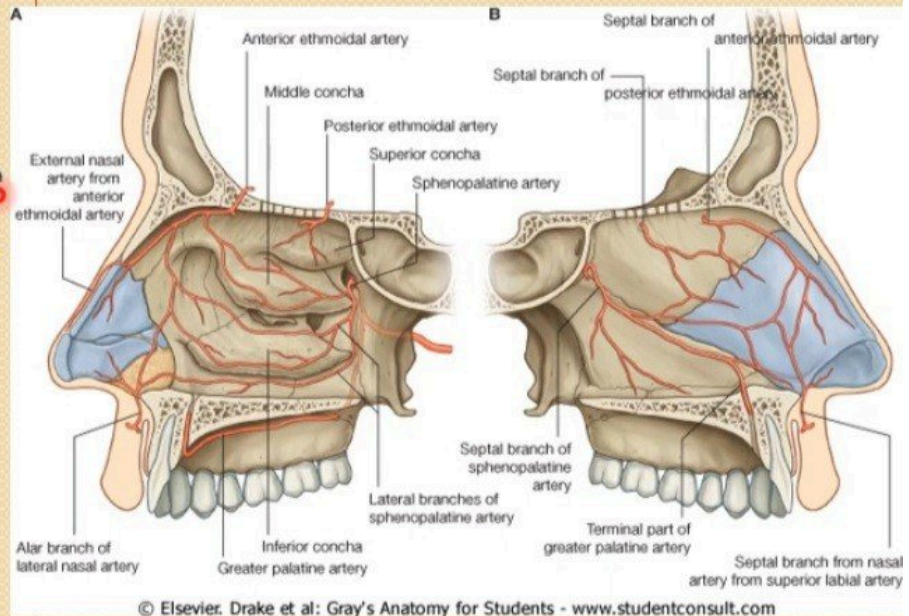
Arterial Supply:

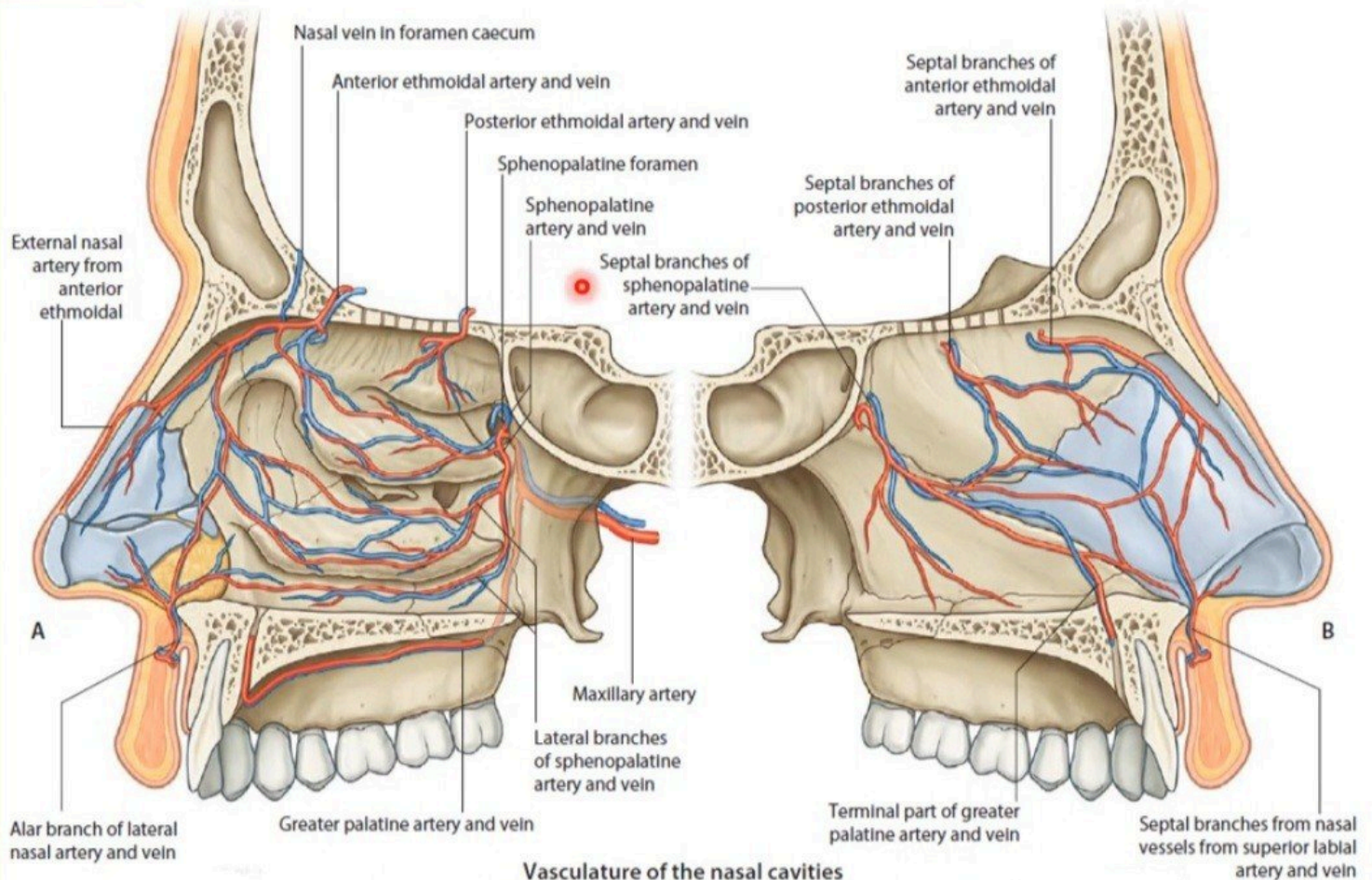
- Branches of the maxillary, facial & ophthalmic arteries.
- The arteries make a rich anastomosis in the region of the vestibule and anterior portion of the septum.

Venous Drainage:

- drain into the facial, ophthalmic, and spheno-palatine veins.

BLOODY SUPPLY



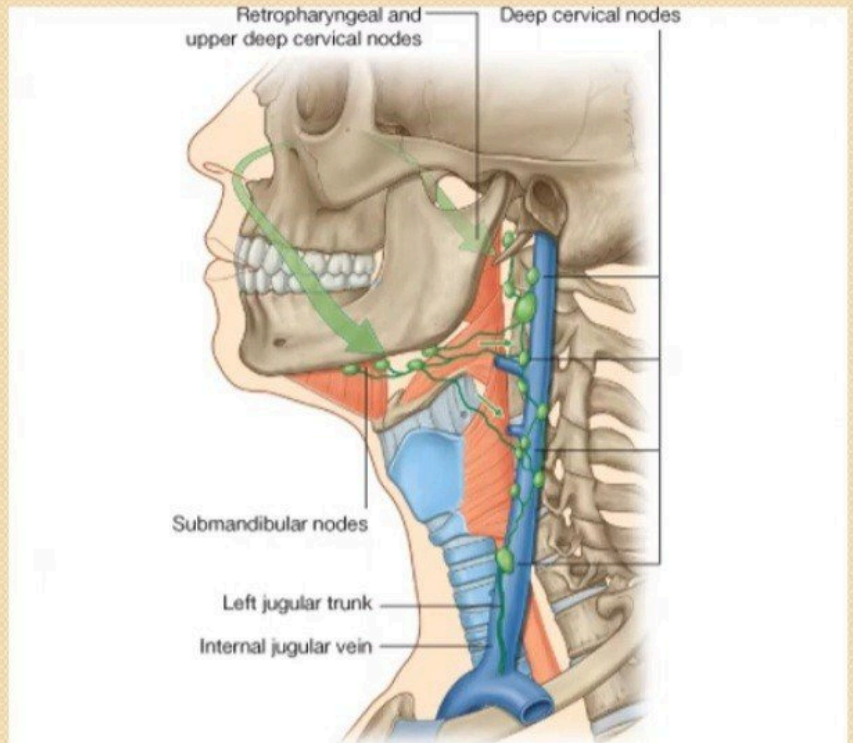


Vasculature of the nasal cavities
A. Lateral wall of right nasal cavity B. Medial wall of right nasal cavity

LYMPHATIC DRAINAGE

The lymphatics from the:

- **Vestibule** drains into the **submandibular** lymph nodes.
- **Rest of the cavity** drains into the **upper deep cervical** lymph nodes.

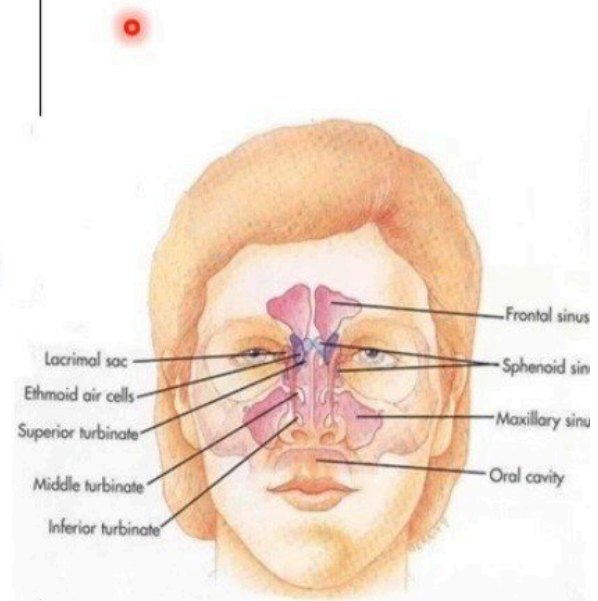


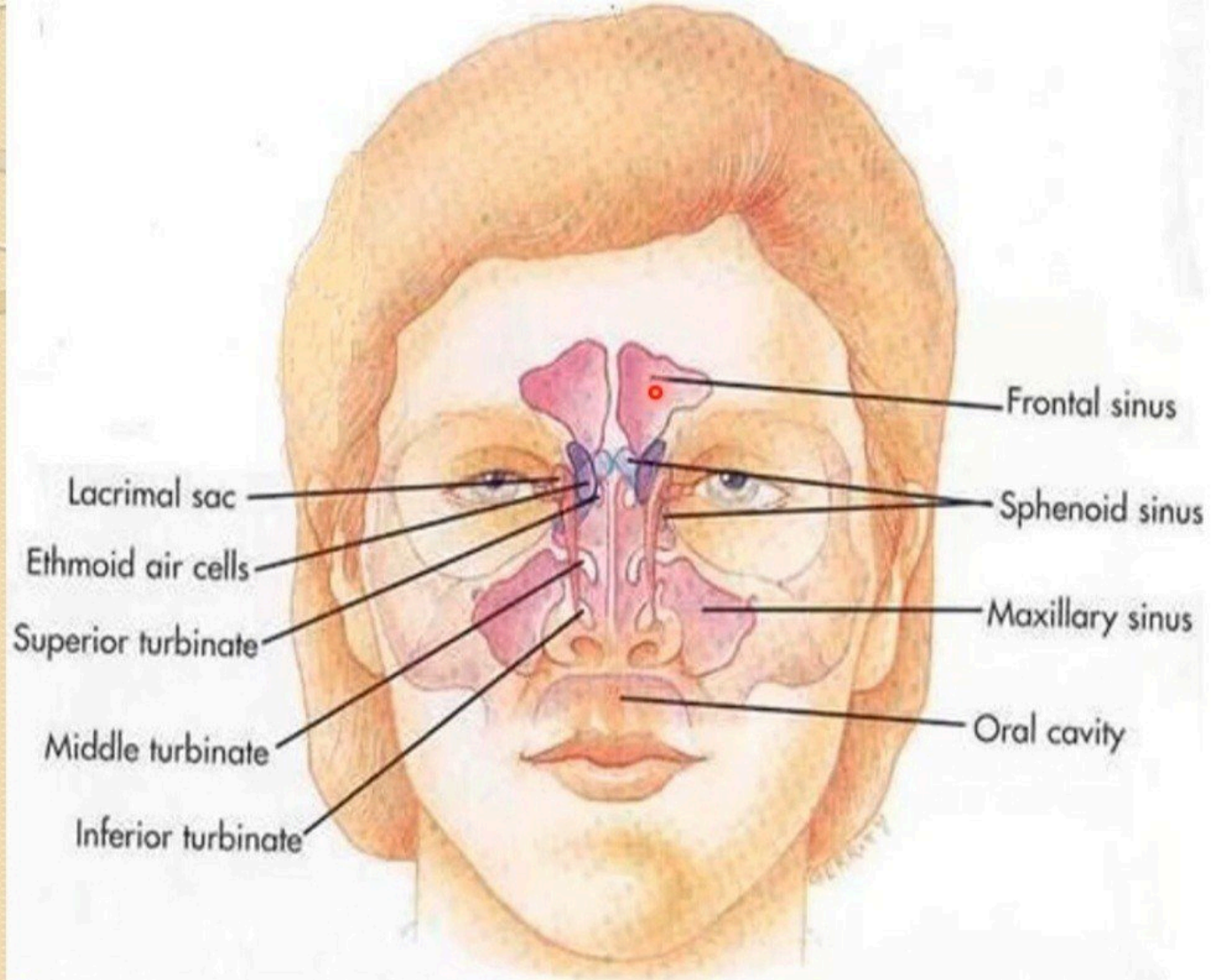
Paranasal Sinuses

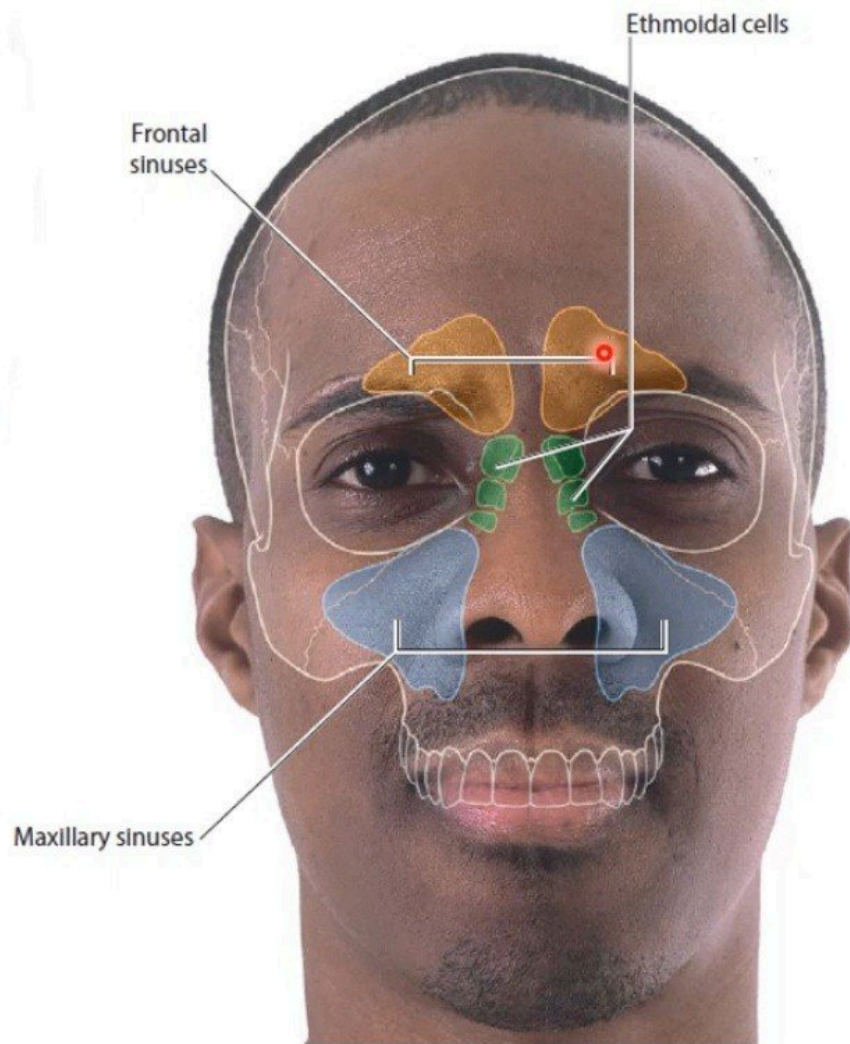
- **Air filled cavities** located in the bones around the nasal cavity:
Ethmoid, Sphenoid, Frontal bones Maxillae.
- **Lined by respiratory mucosa** which is continuous with the mucosa of the nasal cavity.
- **Drain into the nasal cavity.**

Functions

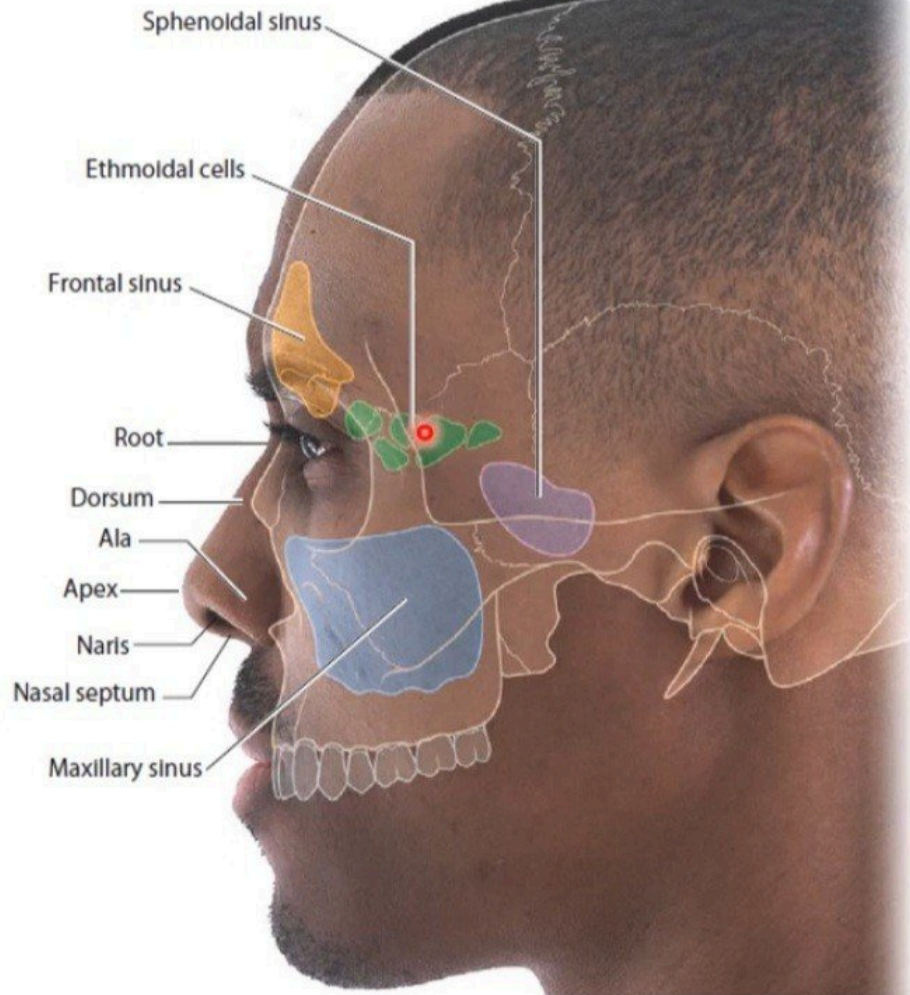
- Lighten the skull.
- Act as resonant chambers for speech.
- **Air conditioning:** The respiratory mucosal lining helps in warming, cleaning and moistening the incoming air.



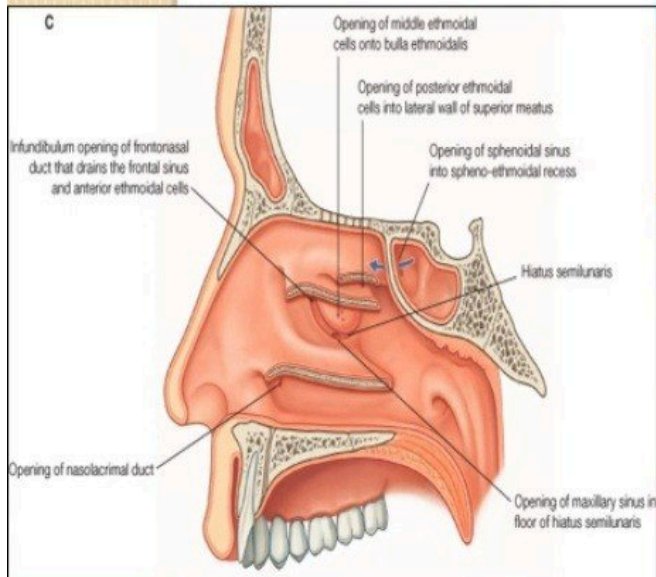




Paranasal sinuses (anterior view)



Surface anatomy of nose and paranasal sinuses (lateral view)



• **Spheno
ethmoidal
recess**

sphenoidal sinus

**Superior
meatus**

posterior ethmoidal sinus

**Middle
meatus**

**middle ethmoidal,
maxillary,
frontal &
the anterior ethmoidal
sinuses**

**Inferior
meatus**

nasolacrimal duct.

C

